

THE ROLE OF INFORMATION SYSTEMS IN PUBLIC REPORTING - A SYSTEMATIC LITERATURE REVIEW

Abstract

Background: *With the advancement of digitalization, health information systems are becoming increasingly essential in healthcare, facilitating the collection, analysis, and transformation of data from various sectors to ensure quality, relevance, and timeliness for informed decision-making. Previous studies indicate that public reporting has, overall, contributed to improvements in healthcare quality and facilitated user choice, including enhanced patient satisfaction and clinical outcomes in pathway theory. However, the reputation pathway, which benefits both organizations and individuals, has been largely overlooked. Additionally, few studies have provided a comprehensive analysis of the effectiveness and limitations of public reporting system.*

Methods: *A literature search was conducted on five electronic databases: PubMed/Medline, PsycInfo, EMBASE, CINAHL, and Web of Science.*

Results: *This review includes findings from a total of 60 research articles. Internet-based public reporting become predominant after year 2015. The purpose for public reporting is categorized into individual -, organization -, and public reporting -related purposes. Policy, organizational perspectives, and behavioral dynamics have influenced the evolution of public reporting. Meanwhile, during the reporting practice, existing systems fail to adequately meet the needs of individual users. Discrepancies in data consistency and accuracy persist across different systems, underscoring the need for a unified data storage standard.*

Conclusions: *This systematic literature review examines public reporting from a global perspective, analyzing its evolution, impact, and future potential. By exploring the relationships between public reporting systems, data components, and key stakeholders, this study provides valuable insights into the effectiveness of public reporting in driving healthcare transparency and quality enhancement. The main contributions to this work are as follows: (1) to provide the up-to-date review of public reporting in health care from a global perspective; (2) identifying key trends, challenges, and gaps in digital public reporting systems, summarizing the current state of knowledge within an integrated framework; and (3) to reduce the complexity of public reporting by offering structure and clarity.*

Keywords: *Health information system, public reporting, digitalization, health quality, systematic literature review.*

1 Introduction

The digitalization of healthcare has long been regarded as a transformative force within the healthcare industry, and the pandemic is thought to have accelerated this process (Golinelli et al., 2020). In this context, health information systems play a crucial role by collecting, analyzing, and ensuring the quality, relevance, and timeliness of data from the health sector and other relevant sources. These systems then transform the data into actionable information that supports health-related decision-making (World Health Organization, 2008). Furthermore, the systematic approach to literature review, a defining characteristic of health technology assessment, serves as a strong foundation for various health assessment methodologies (Fehr et al., 2017). One key approach to enhancing transparency and accountability in healthcare is public reporting, which involves making healthcare quality information publicly accessible to consumers (Kast et al., 2023). Building on this, the primary objective of this study is to provide the up-to-date review of public reporting from the literature on the public reporting in healthcare through a Systematic Literature Review (SLR) (Alavi & Joachimsthaler, 1992; Alavi & Leidner, 2001). By analyzing public reporting across various disciplines, this study not only synthesizes existing knowledge but also offers valuable insights for scholars, policymakers, and practitioners, contributing to the ongoing discourse on healthcare transparency and quality improvement.

Public reporting of comparative information on health care quality of physicians, hospitals, and health plans through “report cards” is hailed as a plausible way to improve health care (Arrow, 1978; Stiglitz, 1989). Report cards may be defined as standardized, publicly released reports on quality of care (Epstein, 1995). Typically, they are used to make comparisons of performance amongst individuals or organizations (Longo et al., 1997), over time or against defined standards of care. In the literature review, we found that current studies have explored a variety of perspectives on public reporting, it shaped by a combination of societal needs and technological advancements (Busse et al., 2019), involving diverse stakeholders (Van Wilder et al., 2021), (Martin M et al., 2013), and policies (Han & Park, 2021). It aims to enhance transparency, facilitate informed decision-making in provider selection, drive continuous quality improvement, and promote accountability in healthcare delivery (Busse et al., 2019).

Previous systematically review has been done mainly in three distinct pathways, in the selection pathway, consumers compare public reporting data and choose high-quality providers over low-quality providers (Berwick et al., 2003). The selection pathway has been systematically studied in terms of patient choice (Behrendt & Groene, 2016), patient satisfaction (Shintani Smith et al., 2021), physician rating website (Lagu et al., 2010). In the change pathway, organizations identify underperforming areas, leading to performance improvement. These pathways are interconnected by providers’ motivation to maintain or increase market share (Berwick et al., 2003). In the change pathway, systematic studies have examined elements such as transparency mechanisms (Timofeyev et al., 2023) and provider-driven changes, including clinical improvements (Behrendt & Groene, 2016; Berger et al., 2013; Campanella et al., 2016), physician’s behavior (Frølich A et al., 2007). Scholars raise the third pathway, public reporting

can negatively affect the public image of a provider or an organization, reputation concerns will therefore motivate providers to protect or improve their public image (Hibbard et al., 2005), thus earn more market share. However, systematically research has rarely been done in the reputation pathway.

Moreover, research has explored the “metric-driven harm” caused by performance measurement systems (Rambur et al., 2013), but some un-intended consequences remain unexplored. For instance, at the individual level, inaccurate reporting information can lead to patient anxiety (Duncanson et al., 2024). Some physicians may avoid high-risk surgical patients to ensure better public evaluations, physicians and hospitals may focus more on achieving specific “target rates” rather than providing personalized care based on clinical judgment and patient needs (Werner & Asch, 2005). At the organizational level, the introduction of public reporting imposes significant financial burdens on hospitals (Lagu et al., 2019). Additionally, reporting bias, data distortion (Martin M et al., 2013), and misuse due to inaccurate outcome data (Lagu et al., 2019). A deeper understanding of these issues is crucial to ensuring that public reporting systems truly enhance health quality rather than introduce new inefficiencies.

This article aims to answer the following questions: (1) What types of public reporting systems and data components of the public reporting are used and studied, (2) What are the origin countries and key user groups of public reporting systems identified in the reviewed literature, (3) What are the main purposes of the reviewed studies on public reporting systems, (4) What are the key findings of the reviewed studies on public reporting systems. Previous systematic literature reviews have largely focused on public reporting from a single perspective. In contrast, this study takes a global perspective, offering a more comprehensive analysis that incorporates multiple viewpoints. By employing a structured framework to examine the alignment between the purpose and outcomes of public reporting, this study advances the existing literature while providing valuable practical implication for researchers, policymakers, and healthcare practitioners.

The remainder of this paper is organized as follows: Section 2 outlines the methodology for planning and conducting the literature review. Section 3 presents the findings corresponding to each research question, summarizing the current state of knowledge within an integrated framework reduce the complexity of public reporting by offering structure and clarity. Section 5 present the finding discusses the implications, limitations, and directions for future research.

2 Methods

2.1 Searching Process of the Systematic Review

A research literature review is defined as "a systematic, explicit, and reproducible approach for identifying, evaluating, and synthesizing the existing body of completed and documented work produced by researchers, scholars, and practitioners" (Fink, 2013), when applied rigorously,

this method fosters comprehensive synthesis, aiming at providing a clear and reliable foundation for future research and practice. The primary aims of such reviews are to identify emerging research areas, facilitate theory development, address gaps in saturated fields, and categorize studies within specific research streams (Okoli, 2015), (Vom Brocke et al., 2015), (Webster & Watson, 2002). This review follows the guidelines proposed by Webster and Watson (Webster & Watson, 2002), (Okoli, 2015). The assessment is complemented by an external evaluation of the outcomes at each juncture.

In the first step, articles were retrieved from five academic databases: PubMed/Medline (National Library of Medicine), PsycInfo (American Psychological Association), EMBASE (Excerpta Medica Database, Elsevier), CINAHL (Cumulative Index to Nursing and Allied Health Literature, EBSCO), and Web of Science (Clarivate Analytics).

Reviewers are constrained to evaluating studies published in languages they can proficiently understand or those accessible via scholarly databases (Fink, 2019). Access to these databases was obtained via the University of Turku facilities, ensuring that the user was logged in to minimize the amount of ignored articles due to access issues. The number of search queries for each database was recorded to verify the accuracy of each subsequent search step. To ensure comprehensive article retrieval and prevent missing items during the search process, the number of search results, downloads from each database, and imports into Endnote software were all monitored. A language restriction was set to English to ensure that the results could be easily queried and referenced by a global audience.

University of Turku librarian was consulted about the search terms. The titles and abstracts of articles were searched using the following terms to identify subject-specific literature for the review: (“healthcare,” “health care,”) and (“public reporting,”). The search strategy for each database is outlined in Table 1, and the detailed explanation of these search strategies is provided. The search string was designed for broad coverage to avoid missing any relevant research. No limitations were placed on publication date or study design; however, studies published in languages other than English were excluded.

Initially we found 2690 publications in total, of which 746 in Pubmed, 563 in WoS, 411 in CINAHL, 830 in Embase and 140 in PsycInfo. Fink recommends electronic tools such as reference managing software to create a “virtual filling cabinet” (Fink, 2019). Duplicate entries were first removed using EndNote software, followed by manual checks for duplicates and title screenings to ensure accuracy.

Database	Search terms
PubMed (746)	("healthcare"[Title/Abstract] OR "health care"[Title/Abstract]) AND "public report*"[Title/Abstract]
PsycInfo (140)	(TI healthcare OR AB healthcare OR TI "health care" OR AB "health care") AND (TI "public report*" OR AB "public report*")
EMBASE (830)	(healthcare OR "health care"):ab,ti AND (("public report*"):ab,ti)
CINAHL (411)	(TI healthcare OR AB healthcare OR TI "health care" OR AB "health care") AND (TI "public report*" OR AB "public report*")
Web of science (563)	(TI=(healthcare OR "health care") OR AB=(healthcare OR "health care")) AND (TI="public report*" OR AB="public report*")

Table 1. Databases and search terms.

2.2 Screening Process of the Systematic Review

In the screening process, first abstracts, keywords, and titles from the five databases of the found articles were screened based on the previously mentioned search terms. Further screening of the 1083 articles was conducted using inclusion and exclusion criteria in a two-step process. In the fourth step, practical screening criteria were applied. The literature review initially screened the titles and abstracts of the 1,083 articles to determine their relevance to public reporting. Selected articles were then subjected to forward and backward citation tracking to ensure the comprehensiveness of the literature search.

After reviewing the titles and abstracts, 993 records were excluded for not matching the inclusion criteria (i.e., studies with the main topic (title/abstract/keywords) containing public report, studies available in English). Our search was confined to English-language publications from 2005 to 2024, as recommended by the Cochrane Rapid Reviews guidance (Garritty et al., 2021). This timeframe was selected based on preliminary searches indicating that the majority of pertinent articles were published during this period.

During the screening process, the titles of the 1,083 retrieved articles were reviewed to determine whether they contained the keyword “public reporting.” If the title is unclear, the

abstract was read to assess whether the study focused on public reporting. Articles that merely mentioned public reporting—such as those recommending attention to public reporting or using public reporting data to examine unrelated research—were also excluded.

Citation chaining was conducted to further reinforce the comprehensiveness of the article search. The inclusion criteria and exclusion criteria for each database is summarized in Table 2. By further not fulfilling the inclusion criteria at the analysis level of full text, 8 articles were excluded, leaving 82 studies.

Inclusion criteria (IC)
IC1: Studies published in English
IC2: Studies with the main topic (title/abstract) includes public report
IC3: Full-texts availability of studies
IC4: Studies published in journals
Exclusion criteria (EC)
EC1: Duplicate studies identified through different searches are eliminated
EC2: Studies whose main topic is not pubic reporting (i.e., even if public reporting is mentioned in the abstract, the article is not primarily about public reporting)
EC3: Full text unavailable
EC4: Studies that are not empirical studies (books, editorials, report, letters) and literature review articles
EC5: Publications from 2005 to 2024

Table 2. *Inclusion and exclusion criteria.*

The types of articles were further screened. The remaining 82 articles were reviewed in full to determine whether they met inclusion criterion 4 (i.e., studies published in journals and conference proceedings). Fifteen literature reviews and 8 non-empirical publications (e.g., books, editorials, reports, letters) were excluded. Notably, the excluded articles were organized into a separate subfolder, categorized according to the specific quality standards they did not fulfill (Okoli, 2015). However, conceptual papers published in peer-reviewed journals were included as they contribute valuable theoretical insights. Reviews and letters were excluded because they did not report original data.

Three articles were excluded from the analysis of public reporting material because their research focused on general discussions of public reporting without providing any specific data relevant to our analysis. At the end, 60 articles were selected as the final sample. Figure 1

illustrates the phases of article retrieval and the number of articles included.

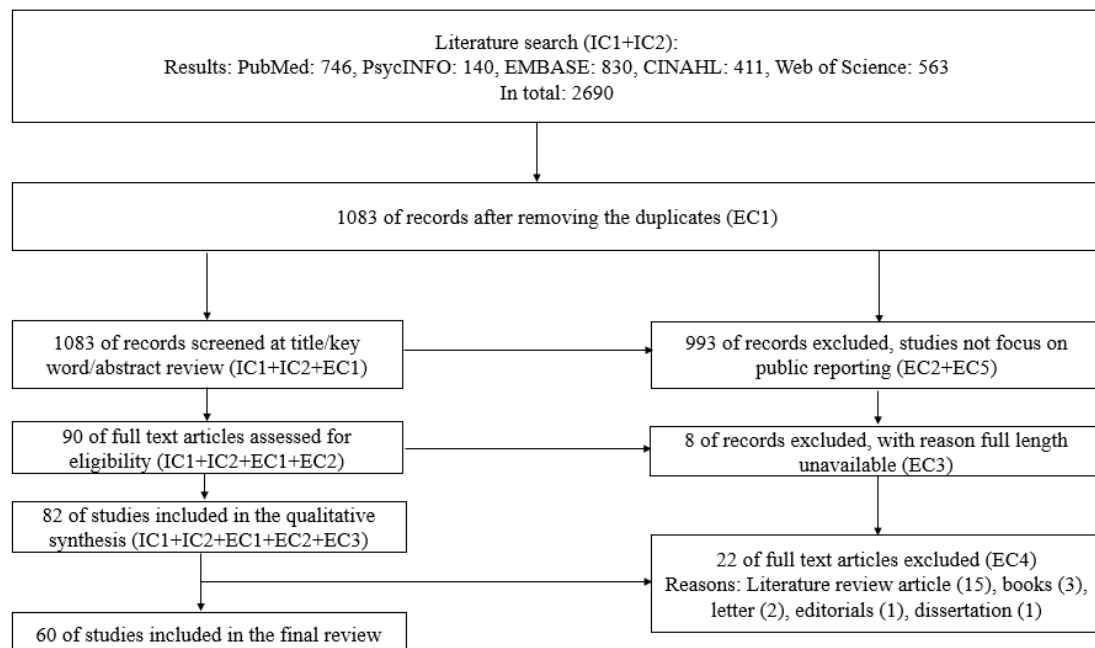


Figure 1. Literature search steps. IC: inclusion criterion; EC: exclusion criterion.

2.3 Data Extraction, Coding and Analysis of the Systematic Review

To analyze the selected articles, definitions, research methods, theoretical frameworks, and study contexts were extracted from the abstract, introduction, methods, and results sections of the reviewed articles. The formulation of research questions was determined through discussions among three collaborators.

In addressing RQ1 (research question 1) and RQ2 (research question 2), the coding process primarily involved keyword queries. For RQ1, an inductive coding approach was applied based on content analysis. In RQ2, the categorization of countries followed an inductive method, while user roles were analyzed using both deductive and inductive approaches. The first deductive approach classified users into "purchaser," "provider," and "policy maker." (Dehmer et al., 2014), whereas the second deductive approach distinguished between individuals and organizations. However, given the diverse user roles, some cases—such as analysts in monitoring systems—did not fit neatly into predefined categories. To avoid subjective biases in classification, an inductive approach was ultimately adopted.

Another challenge in deductive categorization was that certain articles featured multiple roles within the same study. For instance, interviews included both individuals and employees, and in some cases, employees represented both their organizations and themselves. This dual identity introduced potential classification biases, and articles often appeared in multiple categories. To avoid subjective biases in classification, an inductive approach was ultimately adopted.

For RQ3 (research question 3) and RQ4 (research question 4), three coding methods were employed: UTAUT coding, NVivo automated coding, and manual coding. The Unified Theory of Acceptance and Use of Technology (UTAUT) was selected as a coding framework due to its established application in information system research. However, its exploration within public reporting system remains limited. This study, therefore, applies the UTAUT model to analyze the evolution of public reporting in the field of information systems.

In assessing the suitability of these coding methods, several key findings emerged. First, while the UTAUT model provides a structured theoretical framework, it lacks the depth needed to comprehensively address the research questions. Second, although NVivo automated coding ensures objectivity, it proved unsuitable for this study. Given the high volume of complex and nuanced information present in the reviewed literature, NVivo's algorithmic rules lacked the contextual understanding necessary for effective information filtering and in-depth interpretation, limiting its ability to directly address the research questions.

For RQ3 and RQ4, predefined thematic coding was used. The coding framework ensured that no reviewed articles were overlooked, and once all studies were categorized, the extracted purpose statements were systematically mapped to the sections on individuals, organizations, and public reporting systems.

Consequently, this study ultimately employed manual coding. The coding process involved:

1. Looking for the keywords or phrases indicate the answer to the research question from the abstract, discussion and conclusion, then extracting relevant text segments from the reviewed articles.
2. Manual coding has been applied in order to assign preliminary codes to key concepts or themes. after extract the data from the reviewed article to the excel sheet, meanwhile, highlighters have been used to take note on the text in the reviewed article.
3. After the data fit into the theme and can form a coherent pattern, an integrative framework (Huang et al., 2023) has been used in the RQ3 and RQ4 to visualize the alignment between the theme in the purpose and the theme in the result, thus reduce the complexity of public reporting by offering structure and clarity.

Coding was not limited to article titles and abstracts but was derived from the full content of the reviewed studies. Information was summarized and coded into Excel spreadsheets for further analysis. By using this method, the reviewed article can be present as detail as possible. The identified categories were visualized in Figure 3, which informed the results presented in Section 5.

3 Results

3.1 Bibliographical analysis

In the bibliographical analysis we focused on two key indicators: the country that was object of the reporting, and the number of publications per year worldwide.

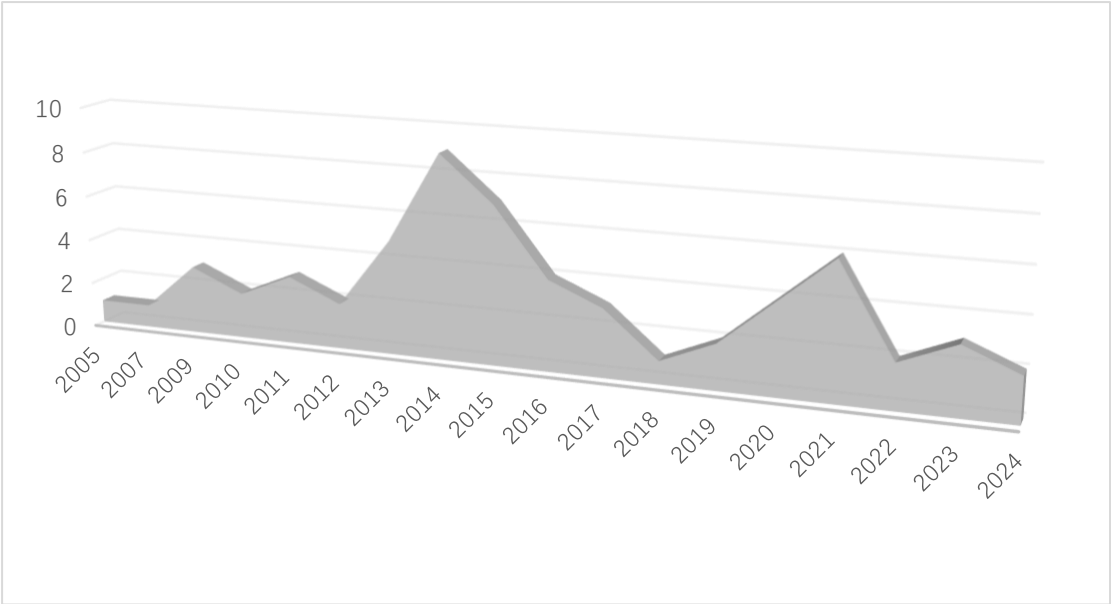


Figure 2. Publications per publication year.

The data reveal two significant peaks in public reporting research, occurring in 2014 and 2021. The first peak in 2014, with nine published studies, marks a high point in public reporting research. However, from 2015 onward, the number of studies steadily declined. As public reporting transitioned to internet-based platforms, the academic focus on traditional reporting formats diminished, contributing to this decline.

The second peak in 2021 reflects the profound impact of global events, particularly the COVID-19 pandemic, on the academic community. The pandemic renewed interest in healthcare systems and public reporting, leading to a resurgence in research activity.

Between 2023 and 2024, fluctuations in research output may be attributed to shifts in research priorities and resource allocation. Additionally, part of this variation may result from incomplete data, as studies from 2024 may still be in progress or awaiting publication. Overall, this trend highlights the evolution of public reporting research and its dynamic relationship with technological advancements and major global events, offering valuable insights for future studies.

As was shown in Table 3, after year 2015 the number of countries whose public reporting was studied are the United States (n=26), China (n=3), Germany (n=3), Europe (n=2). Before year

2015, the country who studied public reporting becomes to the United States (n=16), Germany (n=5), Australia (n=3), China (n=1), Canada (n=1), Belgium (n=1), Korea (n=1).

Country of origin	Frequency	Policy
United States	43	Affordable Care Act (ACA) (Han & Lee, 2021), pay-for-performance programs (Crews et al., 2016), mandatory vaccination policies (Paris et al., 2013), mandatory public reporting system for hospital-associated infections (HAIs) (Edmond & Bearman, 2007).
Germany	5	Since 2005, German hospitals have been required to regularly publish structured quality reports (Emmert & Wiener, 2017).
Australia	3	The use of pay-for-performance (P4P) (Duncanson et al., 2024).
Europe	2	Healthcare-associated infections (HAIs) (Martin M et al., 2013), "patient-centred care" strategies, and pay-for-performance (P4P) approaches (Schlesinger et al., 2015).

Table 3. *Countries of origin in the reviewed articles.*

Public reporting policies play a central role in shaping research contexts. A coding-based analysis of various studies revealed that countries implementing extensive public reporting policies, such as the United States, also produce the highest volume of related research. However, inconsistencies in national policies lead to significant variations in research backgrounds across different countries. Health technology assessment incorporates distinct traditions, practices, resources, and infrastructure, contributing to a multifaceted and evolving landscape (Fehr et al., 2017). To mitigate potential biases arising from these disparities, this study systematically categorizes and analyzes public reporting policies across countries, ensuring a comprehensive and balanced evaluation.

Overall, the United States (n=43) and Germany (n=5) lead not only in the number of policies enacted but also in research activity. Influenced by policy differences, studies in the U.S. primarily focus on enhancing transparency, whereas research in Germany tends to explore the implementation of structured reporting. It is worth noting that while Germany is part of Europe, some studies only refer to "Europe" as a general research focus without specifying individual countries. To maintain objectivity and accuracy in classification, this study preserves the original categorizations used in the reviewed articles, listing Germany separately from studies broadly referring to Europe. This approach ensures the reliability and rigor of the statistical findings.

After 2015, countries like Austria, Belgium, and Korea emerged as countries with public reporting studied, indicating that public reporting has been developing globally. In the study by

Emmert and Wiener (Emmert & Wiener, 2017), a comparison was made between the United States and Germany. Although the research was primarily conducted in Germany, leading to the identification of these two countries in this article. Similarly, the study by Du, Zhang, et al. (Du, Zhang, et al., 2015) evaluated pay-for-performance programs in both the United States and Europe, adding another region to the overall count.

3.1 System types and their input and output

3.1.1 Types of Public Reporting Systems studied

The term "public reporting" encompasses the dissemination of information and reports regarding the quality of care provided by health and long-term care providers, intended for use by patients and other stakeholders. This information is typically accessible through online platforms, brochures, and other printed materials (Kumpunen et al., 2014). Our analysis too indicates that public reporting systems are called with very various names. This also makes their comparison and analysis difficult, as well as finding all relevant articles.

A key distribution goes between digital (computerized) and analog (paper-based) public reporting outputs. By reading the abstracts, introductions, methods and results sections for descriptions, then checking the keywords “internet”, “online”, “web”, “printed”, “paper” and “brochure” in the reviewed article, the majority of public reporting discussed in the reviewed studies are digital (computerized), with the number of 54 (90%). Early studies, particularly those conducted before 2015, focused on analog (paper-based) public reporting, with the number of 3 (5%), which highlights that the Internet has become the predominant medium for studying public reporting. Older studies often centered on analog (paper-based) methods but lacked clarity in their descriptions, whereas newer studies tended to emphasize digital approaches without explicitly detailing their methodologies.

- The key names found in our analysis, that we interpret as being at core public reporting systems, even though they might have other functions too, are presented in Table 4.

Type of System	Frequency	Examples of systems	Articles
Rating Systems	4	• Five-star rating systems • Physician rating websites	(Davila et al., 2024), (Werner et al., 2016), (Emmert et al., 2013), (Emmert & Meier, 2013)
Comparative Reporting Systems	5	• Hospital compare • Home health compare	(Han & Lee, 2021), (Lagu et al., 2019), (Jung et al.,

		• Nursing home compare • Medical hospital compare	2016), (Werner et al., 2009), (Paddock, 2014)
Claims-Based Systems	4	• All-Payer Claims Databases (APCD) • Wisconsin Collaborative for Healthcare Quality (WCHQ) and Wisconsin Health Information Organization (WHIO)	[41], (Han & Lee, 2021), (Crews et al., 2016), (Toussaint et al., 2014)
Patient-Centered Systems	5	• Electronic Health Records (EHR) • Electronic Medical Record (EMR)	(Prang et al., 2021), (Poldrugovac et al., 2022), (Prang et al., 2021), (Tang et al., 2017), (Walters et al., 2014)
Monitoring and prevention Systems	6	• Health care-associated infections (HAIs) • Public Health Automated Case Event Reporting (PACER) • Public health systems	(Martin M et al., 2013), (Hebden, 2012), (Mazor KM et al., 2009), (Edmond & Bearman, 2007), (Mishra et al., 2020), (Chae et al., 2022)
Performance Measurement/Registries	4	• Performance measurement systems • Australia and New Zealand Dialysis and Transplant Registry (ANZDATA) • Canadian Institute for Health Information (CIHI) • National Cardiovascular Data Registry (NCDR)	(Duncanson et al., 2024), (Rambur et al., 2013), (Poldrugovac et al., 2022), (Dehmer et al., 2014)
Public Reporting Systems	7	• Report cards (e.g., hospitals, nursing homes) • Facebook & Yelp • Ascension Health	(Geraedts et al., 2018), (Emmert & Wiener, 2017), (Emmert et al., 2017), (Sander et al., 2015), (Konetzka et al., 2015), (Lagu & Greaves, 2015), (Fakih et al., 2016)

Hospital and Institutional Systems	4	• Hospital Information System (HIS) • National Survey of Substance Abuse Treatment Services (N-SSATS)	(Du, Wang, et al., 2015), (Yang et al., 2014), (Vaughn et al., 2020), (Saghafian & Hopp, 2020)
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Table 4. *Types of Public Reporting Systems in Reviewed Articles.*

The most frequently studied system is the public reporting system (n=7), which includes hospital report cards, provider report cards, and nursing home report cards. Ranked alongside others in second place are the monitoring and prevention system (n=6). Ranked third is the comparative system, which has gained prominence due to the growth of the internet and the expanded use of user-generated data. This trend includes the application of health-related data in various contexts, reflecting the increasing reliance on digital platforms and data-driven decision-making.

3.1.2 Key data sources and output forms for public reporting

As we look at data input components utilized in public reporting systems in the included studies, the most important category is hospital performance data (n=17). As was shown in Table 5, Electronic Health Records (EHR) data list in the second, which align with the patient center system. However, website information, frequently appearing alongside EHR data, ranked third. This finding reflects the increasing dominance of the internet in public reporting, particularly in patient data exposure through comparative systems. While patient-centered systems and website information are frequently referenced, previous studies have highlighted concerns regarding the generalizability of collected data due to the online nature of surveys, potentially excluding individuals without internet access or digital familiarity (Emmert et al., 2013).

Data/Information	Frequency	Examples of data/information
Data		
Quality data	6	Acute myocardial infarction (AMI) (Lagu & Greaves, 2015; Saghafian & Hopp, 2020), survival rate (Walters et al., 2014), clinical notes and imaging studies, such as radiographs (Walters et al., 2014), elective surgery wait times (Prang et al., 2021).
Hospital performance data	17	Healthcare-associated infections (HAIs) data (Han & Park, 2021), 30-day readmission rates (Han & Park, 2021), Percutaneous Coronary Intervention (PCI) mortality rates (Wadhera et al., 2019), infection prevention scores (Duncanson et al., 2024), operating expenses (Han & Lee, 2021), hospital beds (Han & Lee, 2021; Yang et al., 2014), full-time equivalent staff (Han & Lee, 2021), physician information (Geraedts et al., 2018), patient satisfaction scores (Emmert et al., 2017), mortality rates (Crews et al., 2016), antibiotic prescription percentage ranking (Du, Zhang, et al., 2015), chronic condition management (e.g., diabetes care) (Snowden AM et al., 2012; Yonek et al., 2014).
Public database	1	The National Survey of Substance Abuse Treatment Services (N-SSATS) data (Vaughn et al., 2020).
Public health data	3	Healthcare-associated infections (HAIs) data (Fakih et al., 2016; Han & Park, 2021), healthcare resources data (Mishra et al., 2020).
Electronic health records (EHR) data	10	Patient demographics (Tang et al., 2017), prescriber details (name, work unit, time of prescriptions) (Johnson & Talbot, 2011; Martin M et al., 2013).
Healthcare-associated infections (HAI)	3	Healthcare-associated infections (HAI) Surveillance data (Hebden, 2012).
Information		

Website information	7	Comparative quality information (Yonek et al., 2014), rating Information (Emmert & Meier, 2013), quality measures and star ratings for nursing homes (Werner et al., 2016).
Hospital report cards (HRC)	4	Hospital metrics (satisfaction, mortality rates, adherence to care guidelines) (Konetzka et al., 2015).
Patient-Reported information (PRI)	3	Patient outcomes (Lihn et al., 2015; Rambur et al., 2013).
Information on the quality of healthcare providers	1	Risk-adjusted mortality rates (RAMR) for coronary angiography (Sander et al., 2015), patient experience (Toussaint et al., 2014), different information presentation formats (e.g., tables, bar charts) (Sander et al., 2015).
Publicly available performance information	2	Chronic conditions (Shaller et al., 2014).

Table 5. *Data (input) and information (output) components of public reporting in the reviewed articles.*

Information published by public reporting systems consists of processed data, the processing directed at increasing its usefulness (Ackoff, 1989; Totten et al., 2012). However, public reporting faces several challenges across data quality, performance measurement, and consumer engagement. From a data quality perspective, issues include the reliability of administrative claims data and time lags in public reporting, which diminish its relevance for real-time decision-making (Lihn et al., 2015). Potential biases may arise from manipulation or selective participation, further affecting data integrity (Emmert & Meier, 2013).

In terms of performance data, challenges include inconsistencies in presentation formats, leading to difficulties in consumer interpretation (Hafner JM et al., 2011). Many consumers struggle to assess multiple provider performance measures without trusted guidance or simplified presentations (Shaller et al., 2014). Additionally, unintended consequences of public reporting, such as poorly adjusted data influencing provider behavior, may lead to risk-averse practices, such as avoiding high-risk patients to maintain favorable metrics (Rhoads et al., 2009).

Report cards introduce another layer of complexity, as providers may selectively report data or avoid high-risk patients to improve their rankings, resulting in biased samples (Werner & Asch, 2005). Furthermore, discrepancies between reported data and actual quality can influence how report cards are perceived and utilized (Pimentel et al., 2021). Data biases have also been noted, particularly in pain assessments for dementia patients, where ascertainment issues affect reporting accuracy (Konetzka et al., 2015).

Electronic health records, despite their value in public reporting, pose challenges related to harmonization, incomplete data, and inconsistent collection methods (Smith et al., 2023). Public health data, while crucial, faces substantial barriers, including a lack of semantic interoperability that hinders data normalization across sources (Mishra et al., 2020). Variability in reporting methodologies across hospitals and countries further complicates benchmarking efforts (Martin M et al., 2013), (Johnson & Talbot, 2011).

Patient-reported information presents additional obstacles, such as low patient engagement, fragmented data collection, insufficient coordination, and underrepresentation of diverse patient experiences (Lagu & Greaves, 2015). The integration of data from various sources, such as patient narratives, complaints, and patient-reported outcome measures (PROMs), remains a significant challenge (Schlesinger et al., 2015).

Finally, challenges also arise from human factors, including data entry errors and misinterpretation of definitions, necessitating periodic manual corrections that add complexity to the reporting process (Paris et al., 2013). These findings underscore the multifaceted challenges in public reporting, highlighting the need for improved data standardization, risk adjustment, and enhanced usability to ensure the effectiveness of public reporting in healthcare.

3.2 Stakeholders of Public Reporting Systems Identified in the Reviewed Literature

There is an ongoing debate about the effectiveness of existing Health Technology Assessment (HTA) (Fehr et al., 2017). Beyond methodological concerns, discussions emphasize the inclusion and education of key stakeholders and end-users, the adoption of a systematic approach to prioritizing HTA topics, the integration of real-world data, and the establishment of criteria for disinvestment (*International Society for Pharmacoeconomics and Outcomes Research*, 2016). A systematic analysis of user roles within public reporting systems can help identify diverse needs and examine how these variations influence system development.

In the context of public reporting, user roles are traditionally categorized into purchasers, providers, and regulators (Dehmer et al., 2014). However, the emergence of new user roles, seldom considered in previous studies, led to an inductive approach in this section. Notably, the reviewed articles often discuss multiple user roles, with some employing mixed-methods approaches, thereby incorporating multiple users in a single study.

To provide a comprehensive overview of user engagement in public reporting, we have compiled statistical data on user roles based on their representation in the reviewed studies. As presented in Table 6, findings indicate that policy makers (n=33) are the most frequently studied stakeholder group in public reporting research.

Category	Frequency	Examples of stakeholders
Patients	17	Hospital Patients (Jung et al., 2016), Hospital Report Cards (HRC) Users (Emmert & Wiener, 2017), (Sander et al., 2015), Family Members (Sander et al., 2015), (Lihn et al., 2015), (Lagu & Greaves, 2015), (Konetzka et al., 2015), (Du, Wang, et al., 2015), (Shaller et al., 2014), (J. B. Christianson et al., 2010), Elderly Patients with Chronic Conditions (J. Christianson et al., 2014), Community Patients (Jung et al., 2016).
Family member	8	Patient and family member (J. B. Christianson et al., 2010; Du, Wang, et al., 2015; Konetzka et al., 2015; Lagu & Greaves, 2015; Lihn et al., 2015; Mazor KM et al., 2009; Sander et al., 2015; Shaller et al., 2014).
Staff	50	Hospital Administrators (Bayindir & Schreyögg, 2023; Edmond & Bearman, 2007; Ellis et al., 2020; Han & Lee, 2021; Mazor KM et al., 2009; Natale et al., 2011; Poldrugovac et al., 2022; Werner et al., 2009), Infection Prevention Experts (Martin M et al., 2013), (Hebden, 2012), Healthcare IT Specialists (Mishra et al., 2020), Nursing Staff (Konetzka et al., 2015), (Rambur et al., 2013), Staff responsible for evaluating the performance of hospitals (Paddock, 2014).
Analysts	4	Data Analysts (Crews et al., 2016), (Snowden AM et al., 2012), (Hebden, 2012), Public Reporting Developers (Shaller et al., 2014).
Policymakers	33	Public Health Researchers (Smith et al., 2023), (Chae et al., 2022), (Poldrugovac et al., 2022), (Van Wilder et al., 2021), (Han & Lee, 2021), (Vaughn et al., 2020), (Saghafian & Hopp, 2020), (Mishra et al., 2020), (Ellis et al., 2020), (Lagu et al., 2019), (Johnson & Talbot, 2011), (Werner et al., 2009), (Mazor KM et al., 2009), (Edmond & Bearman, 2007), Health Bureau Administrators (Du, Wang, et al., 2015), (Prang et al., 2021).
Organizations	11	Health Insurance Companies (Negele et al., 2023), (Yonek et al., 2014), (Yang et al., 2014), (Toussaint et al., 2014), Professional Organizations for Developing and Evaluating Quality Reports (Dehmer et al., 2014), (Rambur et al., 2013), (Rhoads et al., 2009), Health Department (Yang et al., 2014; Yonek et al., 2014), Infection Control Team (Fakih et al., 2016), Canadian Nursing Center Government Regulatory Agency (Poldrugovac et al., 2022).

Table 6. *Classification of Stakeholders in Reviewed Articles.*

Despite the primary aim of public reporting being to aid users in making healthcare decisions, there is limited research focusing on patients (n=17). However, in our prior research question, patient-related systems were second only to hospital report cards.

It is also noteworthy that, at the individual level, patient family members were included in studies, with findings suggesting that their involvement enhances clinical outcomes. However, system use by this role remains underdeveloped.

The segment of staff involved in monitoring healthcare data was included in studies, consistent with our earlier finding that monitoring and prevention systems receive significant attention and research, with the role of analysts listed in Table 6, aligns with the finding there are new role in the internet-based public reporting definition.

3.3 Key research questions in the articles

The themes within the articles were analyzed using content analysis. This is followed by a presentation of the results. The findings are then summarized in tables for clarity and reference. The original purpose of public reporting was to improve hospital quality and assist patients in making informed healthcare decisions. After investigating all the research questions, we will discuss in the discussion section whether the definition and function of public reporting have diverged from their original intent.

No.	Research question	Examples
RQ1	What types of public reporting systems and data components of the public reporting are used and studied	- What are the typical types of public reporting systems? - What are the key data sources and output forms for public reporting?

		- What types of data are utilized in public reporting?
RQ2	What are the origin countries and key user groups of public reporting systems identified in the reviewed literature	- Which countries conduct the most research on public reporting? - Does national policy influence the volume of public reporting research? - What are the user groups identified in the reviewed studies?
RQ3	What are the main purposes of the reviewed studies on public reporting systems	- Do the reviewed studies share common research purposes? - Are study objectives focused on selection, quality improvement, or reputation pathways? - Do the reviewed articles focus on different user groups?
RQ4	What are the key findings of the reviewed studies on public reporting systems	- Do purpose align with results? - What are the most frequently reported outcomes in public reporting research? - What unresolved challenges remain in public reporting studies?

Table 7. *Classification of Research questions in Reviewed Articles.*

3.4 Main Purposes of the Reviewed Studies

3.4.1 The analysis framework

A theoretical framework to present a holistic view of these factors is developed, which could guide future research in the field (Huang et al., 2023). Purpose resulted in three main themes: individual -related purposes, organization -related purposes, and public reporting -related purposes. For individual -related purposes, we included aspects such as personal behaviors and health disparities. For organization -related purposes, we focused on organizational behaviors and attitudes. For public reporting -related purposes, we examined quality indicators, performance indicators, data utilization, and system usage. Ultimately, we organized the studies into eight themes, which will be discussed in the following section.

3.4.2 Individuals -Related Purposes

Individual Choice

When relates to individual's choice, studies have explored the impact of public reporting on healthcare quality, particularly examining the potential positive and negative effects of report cards on the behaviors of doctors, hospitals, and patients (Werner & Asch, 2005). Other studies aim to understand how public reporting influences individuals' healthcare choices; for example, social media review systems (Lagu & Greaves, 2015) have significantly enhanced individuals' understanding of healthcare performance. Research has also investigated attitudes and preferences related to public reporting data among patients in community living centers (Davila et al., 2024). Additionally, some studies have focused on differences in public reporting preferences between HRC users and non-users (Emmert & Wiener, 2017). Systems such as the nursing home 5-star rating system (Werner et al., 2016), quality ratings for home care institutions (Jung et al., 2016), and physician rating websites (Emmert et al., 2013) have all been examined for their impact on personal healthcare decision-making.

Health Literacy

Additionally, studies have analyzed how different cognitive backgrounds (Emmert et al., 2013) and decision-making contexts (Shaller et al., 2014) affect consumer understanding and decision-making related to public reporting. Factors such as data consistency, report format (text or chart), and the use of confidence intervals have been examined for their influence on consumer comprehension and decision-making (Mazor KM et al., 2009). Research also examined the impact of various presentation formats, such as symbols and charts, to identify methods that improve user comprehension (Sander et al., 2015). Studies have considered the impact of quality reports on the care of patients with advanced dementia (Konetzka et al., 2015), and factors affecting elderly chronic patients' perceptions of physician quality (J. Christianson et al., 2014). Community studies have explored ways to use public reporting to reduce health disparities and improve awareness among minority groups (Yonek et al., 2014).

3.4.3 Organization -Related Purpose

Organization Attitude

Studies on organizational attitudes have often included physicians, as their behaviors and attitudes can represent those of the healthcare institution they work for. Most research has focused on policies and data within public reporting systems. Studies have examined the perspectives of healthcare system stakeholders on the quality policies related to public reporting (Van Wilder et al., 2021), as well as the attitudes of pediatric intensive care (PICU) directors towards benchmarking, public reporting, and pay-for-performance policies (Natale et al., 2011). Physician attitudes toward patient experience data made public online (Lagu et al., 2019), the availability of information, along with perceived value and attitudes of general practitioners to the behavioral intentions (Du, Zhang, et al., 2015), the perspectives of European infection control stakeholders on reporting HAI data (Martin M et al., 2013), and the attitudes of community living centers' (CLC) caregivers toward public reporting systems (Lloyd et al., 2021) (Pimentel et al., 2021) have also been analyzed.

The phenomenon of “metric-driven harm” has been analyzed through case studies to examine its effects on healthcare outcomes (Rambur et al., 2013), while the financial costs and administrative burdens of PCI result reporting programs on hospitals and physicians have been investigated (Wadhera et al., 2019).

Policy

This section explores how transparency within collaborative projects contributes to quality improvement and reduced mortality rates. Studies have offered practical recommendations for enhancing quality competition in Germany (Negele et al., 2023) and examined the impacts of public reporting transparency on patient choice, hospital investment, social welfare, and healthcare market structures (Saghafian & Hopp, 2020), (Lihn et al., 2015). The studies further investigate the foundational assumptions of mandatory public reporting and assess its potential effectiveness in improving hospital quality (Edmond & Bearman, 2007). Other research analyzes transparency mechanisms, such as how the publication of prescription data affects prescribing behaviors (Du, Wang, et al., 2015). Two public reporting programs were compared for their differing impacts on clinical and perceived quality (Han & Park, 2021). Research also examined how public reporting and market competition influence hospitals’ technical efficiency (Han & Lee, 2021) and explored the relationship between patient experience and financial incentives to understand how public policy and healthcare practices can better promote patient-centered outcomes and choice (Schlesinger et al., 2015). Specifically, studies have explored how public reporting mechanisms affect general practitioners’ prescribing behaviors, focusing on factors such as information availability, GP attitudes, perceived value, and perceived risk (Du, Zhang, et al., 2015).

3.4.4 Public Reporting System -Related Purposes

Quality Indicators

Scholars have examined how public reporting influences individual choices in healthcare, finding that quality indicators play a significant role. Patients often consider structured data, such as survival and infection rates, when making choices (Duncanson et al., 2024). Studies also explore the impact of public reporting on quality improvement, particularly regarding process indicators like reperfusion rates and mortality rates for AMI patients (Chae et al., 2022), as well as the importance of quality indicators in selecting hospitals and physicians (Ellis et al., 2020). Other studies focus on how public reporting systems have improved healthcare quality and reduced costs in Wisconsin and California (Toussaint et al., 2014), or whether mandatory

public reporting assumptions effectively enhance hospital quality (Edmond & Bearman, 2007). Research shows that public reporting can support consumer choice and encourage providers to improve care quality (J. B. Christianson et al., 2010) and that it has had positive effects on post-acute care quality in U.S. nursing homes (Werner et al., 2009), cancer surgery wait times (Prang et al., 2021), appropriate antibiotic use in URTI patients (Yang et al., 2014), and flu vaccination rates among healthcare workers (Paris et al., 2013). Moreover, public reporting has contributed to quality improvement in cardiovascular specialty care (Dehmer et al., 2014). Emerging strategies to improve vaccination rates as a quality and safety indicator for healthcare workers are also being explored (Johnson & Talbot, 2011), alongside home healthcare quality improvement efforts (Jung K et al., 2010).

Performance Indicators

Studies have assessed whether performance indicators in long-term care facilities show improvement following public reporting and compared these trends with non-reported indicators (Poldrugovac et al., 2022). Research has also examined the information valued most by referring physicians in hospital quality reports, such as hospital performance and patient experience data, to better meet physicians' needs (Geraedts et al., 2018). Additionally, studies have shown that public reporting of healthcare performance data reveals disparities across insurance plans, encouraging improvement (Snowden AM et al., 2012) (Rhoads et al., 2009). Comparative studies on evaluating provider performance (Paddock, 2014) and the impact of public reporting and pay-for-performance incentives on quality improvement further emphasize how reputational and financial incentives drive providers to elevate care standards. The effectiveness of public reporting data in understanding and enhancing rural Australian hospitals' operational performance has also been evaluated (Lloyd et al., 2021). Other studies investigate how public reporting influences physicians at various performance levels (Tang et al., 2017) and whether public reporting can improve German hospitals' quality without financial incentives (Bayindir & Schreyögg, 2023). Research on staff perceptions of publicly reported performance data highlights its role in internal decision-making and actions (Hafner JM et al., 2011).

Data Usage

Research has explored the feasibility of using state-level electronic health record (EHR) data to measure and reduce health disparities (Smith et al., 2023). Studies also assess consistency in hospital recommendations in Germany's report card system and the causes of ranking inconsistencies (Emmert et al., 2017). Evaluations of claims-based data's compliance with regulatory standards and its applications in hospital quality reporting and pay-for-performance programs have been conducted (Crews et al., 2016). To ensure accuracy and consistency in publicly reported infection data, standardized definitions and monitoring methods are recommended (Hebden, 2012). Infection Composite Scores (ICS) have been developed to help hospitals assess infection control performance over time, identify underperforming areas, and implement improvements (Fakih et al., 2016). The use of data from the National Survey of Substance Abuse Treatment Services (N-SSATS) has been studied for its ability to capture additional information relevant to accessibility and service coordination (Vaughn et al., 2020).

System Usage

The role of the Public Health Automated Case Event Reporting (PACER) system and FHIR interoperability standards in improving public health reporting, especially during rapid infectious disease outbreaks, has been examined (Mishra et al., 2020). Studies have developed systems to track critical health outcome indicators in head and neck cancer treatment to enhance clinical practice and public reporting (Walters et al., 2014). When analyzing the rating distribution on the public reporting website Jameda, researchers examined how patient and physician characteristics influence overall scores (Emmert & Meier, 2013).

The reviewed articles primarily focus on specific aspects of public reporting. From the individual perspective, studies have addressed patient satisfaction (Derks et al., 2024). From the organizational perspective, research has examined topics such as transparency mechanisms (Duncanson et al., 2024), clinical improvements (Derks et al., 2024), health disparities (Wadhera et al., 2019), incentive policies (Kurtzman ET et al., 2008), quality policies (Van Wilder et al., 2021), racial disparities (Negele et al., 2023), infection indicators (Emmert & Wiener, 2017) and incentive structures (Osińska et al., 2022). Studies on public reporting systems have focused on system development (Theis et al., 2017), physician rating websites (Davila et al., 2024), website information display (Vaughn et al., 2020), and information presentation formats (Lloyd et al., 2021), (Kolb et al., 2018). Other areas of focus include quality indicators (Shintani Smith et al., 2021), and data accuracy (Hsu et al., 2019), data consistency (Berger et al., 2013), (Derks et al., 2024). However, few studies have attempted to provide a comprehensive view of the development of public reporting in healthcare.

Main Purpose	Category	Examples
What are the main purposes of the reviewed studies on public reporting systems	Individual	☐ Presentation formats that enhance user comprehension ☐ Influence of physician and patient characteristics on overall ratings
	Organization	☐ Physicians' perspectives on different types of websites ☐ Actual performance of healthcare providers ☐ Metric-driven harm in healthcare assessment
	Public reporting system	☐ Interoperability standards to improve public health reporting, particularly during rapid disease outbreaks ☐ Accuracy and consistency of infection data ☐ Quality indicators

Table 8. *Classification of the main purposes of the Reviewed Articles.*

3.5 Consequences of the reviewed article in public reporting

Result was classified to the formation of three theme, purposes related to individuals, purposes related to organization and purposes related to public reporting system. For purposes related to individuals, we included evaluation tools and health literacy, while purposes related to organization involved organizational attitudes, regional disparities, and adverse impacts. Purposes related to public reporting system focused on quality indicators and standardized frameworks. Ultimately, we categorized the studies into seven themes discussed in the following section.

3.5.1 Individual -Related consequence

Consumer Decision-Making

Studies indicate a correlation between hospital evaluations and traditional quality metrics. Highly rated hospitals on social media often exhibit better quality metrics, though this correlation is not absolute (Lagu & Greaves, 2015). Low consistency in rankings among various report cards has also been observed (Emmert et al., 2017). Research suggests that physician rating websites significantly influence patients' physician selection, and this impact is likely to grow as public awareness increases (Emmert et al., 2013). The star rating system notably affects nursing homes' market shares, with consumers more likely to choose highly rated facilities (Werner et al., 2016). Similarly, public reporting influences consumer choices, with higher-rated home care facilities being more frequently chosen after the introduction of public

reporting (Jung et al., 2016). Ratings for female doctors tend to be higher than for their male counterparts, and older patients and those with private health insurance generally provide higher ratings (Emmert & Meier, 2013).

Health Literacy

Future public reporting developers should focus on user experience by creating reports tailored to different user groups (Shaller et al., 2014). Specific measures can increase minority awareness of public reporting (Yonek et al., 2014). Perceptions of public reporting are influenced by factors such as educational level, health status, and trust (J. Christianson et al., 2014). Patients with advanced dementia tend to perform poorly on specific quality indicators, particularly those related to functional decline and weight loss (Konetzka et al., 2015). MHCP patients generally score lower on quality indicators, with significant gaps persisting despite some improvements (Snowden AM et al., 2012). Results also indicate that users with no prior exposure to hospital report cards show a stronger intention to use them (Emmert & Wiener, 2017).

3.5.2 Organization -Related consequence

Stakeholder Perceptions

The availability of information, along with perceived value and attitudes of general practitioners, significantly impacts behavioral intentions, while perceived risk does not (Du, Zhang, et al., 2015). Acute care hospitals in Flanders have made substantial investments in quality improvement policies, with most hospitals completing initial certifications, publicly reporting quality indicators, and participating in pay-for-performance programs. Despite these efforts, stakeholders hold varying opinions on the effectiveness of these policies (Van Wilder et al., 2021).

Across Europe, public reporting of healthcare-associated infection (HAI) data varies significantly. Experts in countries with established reporting systems generally support public reporting, while those in countries without such systems express reservations. Many experts advocate for reporting structure and process indicators over infection rates, citing concerns about data misinterpretation and biases (Martin M et al., 2013). Providers also worry that transparency could negatively affect their reputation and market share (Lihn et al., 2015).

Physicians often view online quality data on online resources with skepticism, doubting its accuracy. Only 2% consider Medicare's public reporting site highly accurate, and over half believe the data is not helpful for patients choosing doctors. They prefer foundational

information, such as board certifications and clinical interests, over performance-based quality metrics or patient reviews (Lagu et al., 2019). Studies suggest that hospital report cards should emphasize physician and patient experiences to better support consultations. This implies that presenting hospital quality data requires adjustments to maximize its utility in clinical decision-making (Geraedts et al., 2018). The availability of information and the perceived value and attitudes of general practitioners significantly influence their behavioral intentions, whereas perceived risk does not exert a significant effect (Du, Zhang, et al., 2015).

Disparities

Evidence suggests that public reporting has had a significant impact on performance improvement (Poldrugovac et al., 2022). Public reporting transparency policies can enhance social welfare by reducing inequalities and encouraging hospital investments in quality improvement (Saghafian & Hopp, 2020). Transparency efforts have also improved prescribing behaviors and appropriate drug use (Du, Wang, et al., 2015). The NPC-QIC initiative has achieved significant advancements in reducing infant mortality, improving nutrition management, and optimizing care processes (Lihn et al., 2015). In Wisconsin and California, public reporting systems have significantly improved healthcare service quality while reducing costs (Toussaint et al., 2014). However, while higher-performing hospitals have benefited from public reporting, lower-performing hospitals have not shown similar improvements (Bayindir & Schreyögg, 2023).

Studies have also noted variation in improvements across institutions, with some metrics showing significant advancement while others have lagged (Jung K et al., 2010), performance data from public reporting (Lloyd et al., 2021). In one survey, 73% of PICU medical directors believed that benchmarking could improve patient outcomes (Natale et al., 2011). The impact of transparency is more pronounced for younger, wealthier, or urban patients compared to rural or older patients (Saghafian & Hopp, 2020). Given the complex effects of public reporting on hospital efficiency across different market environments, policymakers are advised to consider market dynamics to achieve optimal healthcare outcomes (Han & Lee, 2021). Studies also show that while some public reporting programs have improved clinical quality in specific healthcare domains, others primarily affect public perception rather than actual quality outcomes (Han & Park, 2021). In specific cases, such as elective surgery for lung cancer, waiting times have slightly increased (Prang et al., 2021).

Un-intended consequence

Public reporting, while promoting transparency and accountability, can also result in unintended consequences, often referred to as “metric-driven harm” (Rambur et al., 2013). Patients generally support the public reporting of performance data but report increased anxiety, especially when they cannot change their healthcare provider (Duncanson et al., 2024). They prefer data to be presented positively, accompanied by explanations and educational resources to aid understanding and decision-making. Trust in their healthcare team remains paramount, with many patients relying more on physician advice than on statistical data (Duncanson et al., 2024).

In Germany, quality competition has improved healthcare standards, with quality measurement and public reporting recognized as key drivers of these improvements (Negele et al., 2023). Many studies recommend that regulators prioritize incentives for hospitals to improve quality, considering interventions that encourage hospitals to enhance their services (Saghafian & Hopp, 2020). However, despite the transparency and accountability that public reporting promotes, unintended negative outcomes may arise. For instance, some physicians may avoid high-risk patients to improve their reported performance metrics. Furthermore, doctors and hospitals may focus more on meeting specific “target rates” rather than providing individualized care based on clinical judgment and patient needs. While some studies suggest that public reporting has contributed to lower cardiac surgery mortality rates, others find evidence that some physicians may refuse high-risk cases to maintain favorable evaluations. Additionally, while transparency has generally improved healthcare quality, studies reveal its impact can be limited, and in certain cases, it may even inadvertently reduce care quality (Werner & Asch, 2005).

Financial burdens also accompany public reporting requirements (Wadhera et al., 2019). For example, hospitals performing percutaneous coronary interventions (PCI) face substantial costs, with catheterization labs incurring annual expenses ranging from \$100,000 to \$200,000, and in some cases, even higher (Lagu et al., 2019). Despite these challenges, combining incentive-based programs with patient experience reports and outcomes has shown promise in improving patient-centered care and delivering health outcomes that patients value (Schlesinger et al., 2015).

3.5.3 Public Reporting System -Related consequence

Quality Indicator

Public reporting has been shown to positively impact quality indicators and process measures, such as reperfusion rates and AMI mortality (Chae et al., 2022), and reduce antibiotic prescription rates (Tang et al., 2017; Yang et al., 2014). Meanwhile, CLC quality improvements (Davila et al., 2024; Pimentel et al., 2021), vaccination rates (Paris et al., 2013), quality of care metrics (Hafner JM et al., 2011; Werner et al., 2009), infection control measures (Edmond & Bearman, 2007; Fakhri et al., 2016), and cardiovascular care quality (Dehmer et al., 2014) have all improved following public reporting implementation. However, public reporting systems may require adjustments to focus on factors patients find meaningful, such as hospital reputation and physician expertise, to increase their usage and impact (Ellis et al., 2020).

Standardized Framework

Public reporting’s influence on quality improvement has been limited, especially during the late 1990s, as third-party payers, patients, and referring physicians made little use of publicly available data for clinical decisions (Rhoads et al., 2009). A unified data processing framework could address these issues by providing an effective approach to measuring and monitoring health disparities (Smith et al., 2023). Collaborative efforts between public health agencies and

healthcare systems have enabled faster case information collection, reduced human errors, and ensured data privacy and accuracy (Mishra et al., 2020). Future systems should support automated, structured clinical data entry to effectively track and report treatment outcomes, thereby enhancing patient care and supporting a value-based healthcare system (Walters et al., 2014). Significant discrepancies in monitoring practices across institutions suggest the need for standardized definitions and automated systems to improve data accuracy and consistency (Hebden, 2012). Standardized and validated reporting methods are also essential for improving report credibility and applicability (Johnson & Talbot, 2011) (Crews et al., 2016). Integration of patient experience data and real-time information is increasingly considered essential for the future of public reporting (J. B. Christianson et al., 2010). A new survey framework has been proposed to guide data collection and improve services in the future (Vaughn et al., 2020).

The format of data presentation in public reports, such as using charts or text and including confidence intervals, does not significantly impact consumer understanding (Mazor KM et al., 2009). However, providing detailed explanations within public reporting data enhances individuals' ability to comprehend and make informed choices (Duncanson et al., 2024). Supporting consumers in better understanding performance data is essential. Seven elements of information presentation can improve user comprehension, such as symbol-annotated bar charts, clear labels indicating high and low values relative to hospital performance, and rankings based on performance (Sander et al., 2015).

3.6 A Framework to summarize the research tradition of Public Reporting

Policy maker makes the policy, organization implement the policy in the health care, individual as an information receiver can transfer signal to develop the public reporting, public reporting as a bridge connects the policy maker and the information receiver. This study explores pathway theory, highlighting that most research has focused on the selection and improvement pathways. Although Hibbard (Hibbard et al., 2005) introduced the reputation pathway, which can enhance corporate image and competitiveness, this pathway has seldom been studied in public reporting. From an information systems perspective, the system itself is a key entity. Therefore, this paper introduces the public reporting system as a third entity to examine its role in public reporting.

By analyzing the perspectives of individuals, organizations, and public reporting systems, the study investigates the development of the reputation pathway. The constructed framework diagram systematically elucidates the interactions among these three entities. Additionally, given the discrepancies between the purposes and results in the reviewed articles, this paper systematically explores and reports on this structure.

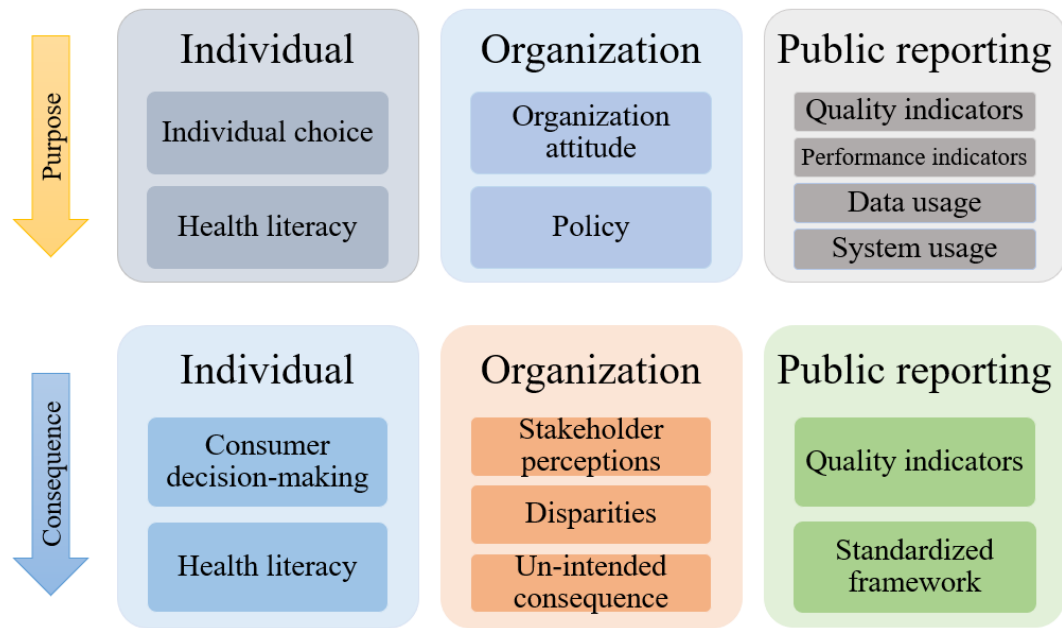


Figure 3. A framework for understanding purpose and results in health care.

From the alignment between the purpose and the result, the evaluated research demonstrated that individual factors can impact public reporting performance. Second, a number of examined articles conclude that public reporting performance can be impacted by health literacy, which is consistent with the health literacy theory (Busse et al., 2019). Third, a number of evaluated research conclude that organizational attitudes impact public reporting performance; yet, different physicians have differing opinions about the data from public reporting. European physicians mostly use internet sources to meet their information demands since healthcare professionals are increasingly needing digital health literacy abilities to incorporate new knowledge into their practice (Demergazzi et al., 2020), (Slevin et al., 2020).

When it comes to the purpose is not aligned with the result, first, public reporting is provided for consumers to help individual choice, result comes to the individual factor affect individual choice, the reviewed articles emphasize the importance of including patient experience reports and physician experience reports, reports should also place greater emphasis on factors that patient consider significant, such as hospital reputation and physician expertise (Ellis et al., 2020). Second, after the publish of the policy, the improvement from the organization differs based on the organization difference. Third, the evaluated article highlights automated and standardized systems. By reducing human error and creating the data standard, data accuracy and consistency may be ensured. Consistent results further emphasize the significance of information presentation, demonstrating that data presentations have a major influence on user comprehension.

While public reporting plays a crucial role in improving healthcare quality, it also leads to unintended consequences in three key areas. First, data-related challenges arise, as public reporting can inadvertently cause anxiety among patients due to the way information is presented. Issues such as data standardization and information clarity should consider individual factors and health literacy to ensure that reporting is both accessible and informative.

Second, clinical decision-making may be influenced by public reporting metrics. Physicians and hospitals may prioritize achieving specific “target rates” over delivering personalized care tailored to individual patient needs. This shift in focus could compromise the quality of patient-centered treatment. However, integrating incentive-based programs with patient experience reports has shown potential in enhancing patient-centered care and improving health outcomes (Schlesinger et al., 2015).

Third, financial burdens are an additional consequence. Compliance with public reporting requirements often imposes significant costs on healthcare institutions. This can explain why public reporting can improve reputation and thus increase its market share, but few research has focus on this pathway.

3.7 A Framework to summarize consumer decision making on public reporting

The rise of social media has significantly reshaped how patients access and evaluate healthcare information. Previous research suggests that patients tend to trust social media ratings more than government websites when assessing healthcare providers (J. B. Christianson et al., 2010; Lagu & Greaves, 2015). The primary reason for this preference is that social media reviews are perceived as being based on “real patient experiences”, allowing for unfiltered, emotional expressions from individuals who have actually undergone treatment (Lagu & Greaves, 2015). Platforms such as Facebook and Yelp provide spaces where patients can freely share their experiences, fostering a sense of authenticity and relatability that government-run quality reports often lack. However, some scholars question the objectivity of social media ratings, arguing that these platforms are susceptible to bias from extreme opinions (Sander et al., 2015). For instance, Yelp reviews tend to attract patients who are either extremely satisfied or extremely dissatisfied, leading to potential distortions in the overall perception of healthcare quality.

To further examine how patients engage with public reports and healthcare reviews, this study analyzes 21 articles that explore the role of public reporting from the perspective of healthcare purchasers. The framework used in this analysis is based on The Consumer Choice Model, initially introduced by (Faber et al., 2009) and further elaborated upon by Busse et al. (Busse et al., 2019). This model provides a structured approach to understanding the different stages consumers go through when making healthcare decisions. It is particularly relevant in today’s digital era, where smartphone adoption is widespread, yet individuals interact with different screens, platforms, and information sources, leading to varying degrees of exposure to and trust in healthcare information.

The Consumer Choice Model outlines four key stages in consumer decision-making regarding healthcare information. The first stage is awareness, where individuals must first recognize the existence of health-related information. In the modern digital landscape, while everyone has access to the internet, the way they engage with healthcare information varies depending on their preferred sources and level of exposure. Some consumers may encounter government health reports, while others primarily rely on social media, online reviews, or word-of-mouth recommendations. The second stage is comprehension, in which consumers must interpret and make sense of the health information they have encountered. The way individuals consume and process health data differs across populations, influenced by education levels, digital literacy, and personal experiences. The third stage is trust, a critical filtering process where consumers assess the credibility of the information before integrating it into their decision-making. This stage is particularly significant given the skepticism surrounding both government reports and social media ratings, as different consumers place varying degrees of trust in these sources. Finally, the fourth stage is behavioral change, where consumers translate trusted information into actionable decisions regarding their healthcare choices. At this stage, users rely on

previously validated and trusted information to guide their healthcare behaviors, whether it be selecting a hospital, choosing a physician, or deciding on a treatment plan.

In the Awareness stage, the extent to which patients recognize and acknowledge public reports largely depends on their health literacy and education level. Studies indicate that individuals with higher education and better health literacy are more likely to be aware of and utilize public reports (J. Christianson et al., 2014; Du, Wang, et al., 2015; Shaller et al., 2014). However, many patients and their families have limited knowledge of Hospital Report Cards (HRC) due to insufficient promotion efforts, leading to a lack of awareness regarding prescription ranking information (Du, Wang, et al., 2015; Edmond & Bearman, 2007; Emmert & Meier, 2013; Jung et al., 2016; Konetzka et al., 2015; Rhoads et al., 2009; Sander et al., 2015; Shaller et al., 2014; Yang et al., 2014). Furthermore, awareness levels differ significantly among demographic groups. Education level, health literacy, race, and overall health status have been found to influence knowledge of public reporting (J. Christianson et al., 2014; Du, Wang, et al., 2015; Shaller et al., 2014). Women, widowed individuals, and those enrolled in national health insurance (SHI) exhibit higher awareness of physician rating websites (Emmert et al., 2013), whereas older adults (65 and above) show low awareness, with only 13% knowing about physician quality information (J. Christianson et al., 2014). Nonetheless, older individuals and those with poor health conditions tend to seek more physician quality data when making healthcare decisions (J. Christianson et al., 2014). Effective promotion of public reports requires targeting key decision-making moments, such as changing doctors, receiving a chronic disease diagnosis, or preparing for surgery (Shaller et al., 2014). Moreover, access channels play a crucial role in awareness levels. Patients who trust the internet as an information source are more likely to be informed about public reports (J. Christianson et al., 2014). The rise of social media has also provided patients with new avenues for accessing hospital quality information through platforms like Facebook and Yelp, which are increasingly preferred over government websites (Lagu & Greaves, 2015).

In the Knowledge stage, comprehension of public reports is hindered by the complexity of quality indicators and the lack of intuitive data presentation (J. B. Christianson et al., 2010; Duncanson et al., 2024; Mazor KM et al., 2009; Shaller et al., 2014). This challenge is particularly pronounced when indicators are overly technical or lack sufficient explanation (J. B. Christianson et al., 2010; Lagu & Greaves, 2015; Werner & Asch, 2005). Research suggests that enhancing data visualization can significantly improve patient understanding. For instance, bar charts with symbols enhance comprehension, whereas tables with word labels tend to be more difficult to interpret (Sander et al., 2015). In this context, healthcare providers and physicians can act as “translators”, helping patients understand the meaning of report data (Shaller et al., 2014). Additionally, the availability of visualization tools further supports comprehension. While the WHIO database provides quality and cost data for 920 medical scenarios, its effectiveness is limited due to the lack of interactive visual tools (Toussaint et al., 2014). Implementing star ratings and user-generated reviews has been found to improve patient

engagement and interpretation of hospital and physician performance metrics (Emmert & Meier, 2013; Lagu & Greaves, 2015; Walters et al., 2014). Therefore, optimizing data presentation and accessibility can facilitate accurate patient decision-making and enhance the overall impact of public reports.

In the Attitude stage, both physicians and patients express skepticism toward public reporting. While physicians are generally aware of public reports, some believe that patients do not use or benefit from these reports (Du, Wang, et al., 2015). Many hospitals and healthcare professionals remain skeptical about PR and PFP programs, viewing them as potentially misleading for patients (Rhoads et al., 2009). Additionally, some physicians argue that public reporting interferes with clinical decision-making, as hospital rankings may be based on incomplete or misinterpreted data (Du, Wang, et al., 2015; Edmond & Bearman, 2007; Toussaint et al., 2014; Yang et al., 2014). Although public reporting is regarded as a valuable tool for improving healthcare quality, concerns persist regarding misuse and misinterpretation of data (Toussaint et al., 2014). On the patient side, trust in public reports remains low, as many fear that hospital ratings can be manipulated (Edmond & Bearman, 2007; Emmert & Wiener, 2017; Jung et al., 2016; Werner & Asch, 2005). Instead of relying on publicly available data, patients tend to place greater trust in physician recommendations and hospital reputation (Edmond & Bearman, 2007; Mazor KM et al., 2009; Sander et al., 2015; Walters et al., 2014). Moreover, individuals with lower information literacy, as well as those who are less engaged in digital health resources, often depend on physicians and social workers for guidance in healthcare decision-making (Edmond & Bearman, 2007; Jung et al., 2016; Konetzka et al., 2015; Mazor KM et al., 2009; Sander et al., 2015; Walters et al., 2014; Werner & Asch, 2005). This highlights the ongoing challenges in building public confidence in healthcare quality reporting and suggests the need for greater transparency and physician involvement in explaining report data to patients.

In the Behavior stage, despite the availability of public reports, most patients continue to rely on physician recommendations rather than quality rankings when selecting healthcare providers. The limited impact of public reports is primarily due to constraints in healthcare options, as many patients lack the ability to switch providers freely (Duncanson et al., 2024; Jung et al., 2016; Mazor KM et al., 2009; Werner & Asch, 2005). Consequently, only a small proportion of patients alter their healthcare decisions based on public reports, with most continuing to select familiar and trusted hospitals (Duncanson et al., 2024; Jung et al., 2016; Mazor KM et al., 2009; Werner & Asch, 2005). However, at the organizational level, public reports are beginning to gain influence, particularly in the management of chronic diseases (Toussaint et al., 2014). As healthcare systems continue to evolve, data-driven decision-making may play a larger role in shaping hospital practices, yet its direct impact on individual patient behavior remains limited. Moving forward, efforts to increase patient engagement with public reports

should focus on enhancing accessibility, improving comprehension, and integrating report data into key healthcare decision-making moments.

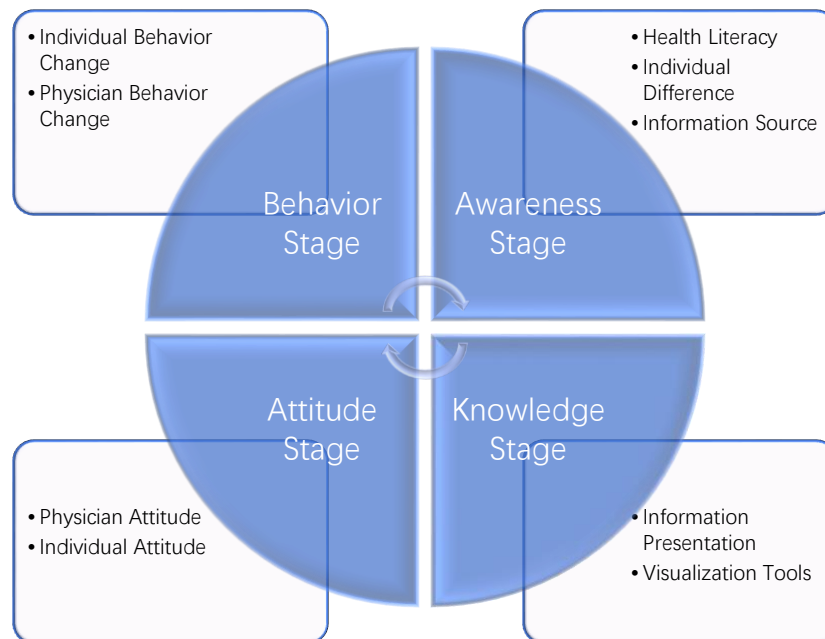


Figure 4. A framework for understanding information processing in public reporting.

4 Discussion

Traditionally, public reporting in healthcare relied on analog (paper-based). However, since 2015, there has been a significant shift towards digital (computerized). Globally, patient-centered public reporting systems incorporating user-generated data have become prevalent. The term 'user' now encompasses not only patients, providers, and regulators but also analysts involved in monitoring and prevention systems, reflecting the expanded scope due to digitalization.

This study systematically identified and reviewed 60 articles on public reporting in health care. Over time, public reporting has garnered increasing interest globally, with the internet become the predominant way of developing public reporting, patient-centered system and website information has been studied frequently, meanwhile, monitoring and prevention system and the new role of analysis, public reporting gained broad definition. Based on the findings, we will propose future research agendas and discuss research contributions and limitations in the following.

4.1 Future Agenda

4.1.1 Conceptual Agenda

Future research should prioritize developing a comprehensive and interdisciplinary conceptual understanding of public reporting. With the development of the digitalization, the definition of public reporting has expanded. Research in a single discipline led to the fragmentation within the findings in the literature, hindering the ability to compare and integrate findings effectively. Therefore, further research is required across multiple disciplines, particularly in fields such as health sciences, information systems, psychology, and computer science, to advance understanding and address the complexities of this topic.

4.1.2 Thematic Agenda

While previous research has predominantly concentrated on individual -related aspects and organization -related aspects, limited attention has been given to public reporting -related aspects in the healthcare context. Additionally, most studies have primarily examined the motivations behind specific individual side and organization side, with fewer studies addressing the barriers to effective public reporting practices. Moreover, while prior research has explored the outcomes of data usage and standard framework for providing an effective approach to measuring and monitoring health disparities, there remains a notable gap in understanding how data processes function across diverse healthcare contexts to achieve these outcomes. Based on these findings, we propose the following thematic agenda.

First, data serves as a powerful driver of economic growth worldwide (Statista, 2022), however, data equals to information plus redundant data (Li et al., 2024). Among public reporting data, performance data and health records data rank as the most frequently reported categories. However, performance data often suffer from complexity and inconsistent presentation formats, making it difficult for consumers to interpret multiple provider performance metrics—especially in the absence of trusted guidance or simplified visualizations (Shaller et al., 2014). In advancing personal systems, reports should also place greater emphasis on factors that patients consider significant (Ellis et al., 2020).

Second, there is a need to explore information presentation in the public reporting. Research shows that information displays significantly impact user understanding, for example, bar graphs with symbols, clear markers of high and low values, and hospital rankings based on performance have been shown to enhance patient comprehension of public reporting data (Sander et al., 2015). Additionally, publicly reported data, such as infection rates, may be misinterpreted or induce unnecessary patient anxiety, further complicating the effective use of public reporting for informed decision-making (Martin M et al., 2013).

Third, there is a need to explore the stakeholder's attitude and actual behavior of public reporting. As digital media capabilities continue to advance, healthcare professionals increasingly require digital health literacy skills to integrate new information into their practices (Slevin et al., 2020). However, many physicians or experts maintain a skeptical or reserved stance on public reporting and use it less frequently (Van Wilder et al., 2021). With the general improvement of organization after the release of public reporting information, future research is needed from the organization side, since organization is the implement side, sitting between the regulator and individual. Therefore, future research can focus on the stakeholder's attitude and actual behaviors.

Forth, there is a need to explore the analysts as a user group. with RQ 1 come up with the monitoring system, analysts had shown up as a new user group in public reporting. Information accessibility and perceived value significantly influence healthcare professionals' behaviors

(Du, Zhang, et al., 2015). Though European physicians primarily fulfill their information needs through online resources (Demergazzi et al., 2020), they may still rely on other sources for specialized or nuanced information. Monitoring system as new information resources need more study in the future research area.

4.1.3 Technological Agenda

Enhancing system interoperability is another critical focus for technological advancements in public reporting. This can increase data usability, improve operational efficiency, and facilitate rapid data sharing during infectious disease outbreaks, thereby resolving issues of data inconsistency across platforms. Data standardization and automated systems have been frequently cited in research as effective means to enhance data accuracy and consistency (Mishra et al., 2020; Smith et al., 2023), (Johnson & Talbot, 2011), (Hebden, 2012), (Walters et al., 2014), (Vaughn et al., 2020), (J. B. Christianson et al., 2010), (Crews et al., 2016). Disparities in storage systems arise from the use of different structures, such as relational databases, NoSQL databases, and file systems. Establishing a unified data standard across systems is essential, especially given the potential for updates and differences in organizational versus personal system storage and rules. Some research also points to significant variations in internal validation methods across healthcare institutions, supporting this conclusion (Crews et al., 2016). The practice of public reporting data varies greatly across European countries (Duncanson et al., 2024), suggesting that future research could focus on data standardization and calibration.

5 Contribution

5.1 Theory Implications

Health information systems play a crucial role in enhancing patient care (Walters et al., 2014). This study contributes to the field by systematically synthesizing, analyzing, and conceptualizing the current state of research on public reporting systems. First, through a structured explanation of public reporting system, it provides a comprehensive framework that reduces the complexity of public reporting by categorizing it into three key dimensions: individual, organizational, and public reporting structures.

Second, this study addresses a critical gap by bridging the disconnect between research objectives and unintended consequences in public reporting. Existing literature has identified that public reporting can lead to unintended consequences, resulting in users' distrust of the information sources (Li, 2024; Moore et al., 2020; Theis et al., 2017). This study further

explores how data transmission between systems contributes to data discrepancies and provides a system-level theoretical analysis of the unintended consequences of public reporting.

Third, while pathway theory has been extensively developed, the reputation pathway—one of its key components—has received limited attention since 2005. This pathway has the potential to enhance an organization's reputation, thereby increasing market share and fostering a mutually beneficial evolution within the healthcare organization.

Public perception of healthcare quality often diverges from actual clinical performance due to selective reporting and misinformation. By refining the mechanisms through which public reporting influences trust and credibility, healthcare institutions can leverage their reputation not only as a competitive advantage but also as a strategic tool for improving care quality and guiding patient decision-making.

5.2 Practical Implications

First, policymakers should consider market mechanisms when implementing public reporting to ensure optimal healthcare outcomes (Han & Lee, 2021), with incentive structures that particularly address the needs of larger hospitals, as different public reporting programs have varying impacts on clinical quality (Han & Park, 2021). Policy should integrate with market mechanisms to maximize the effectiveness of public reporting systems. The degree of improvement varies significantly (Hafner JM et al., 2011) may be due to increased emphasis on data by hospital leadership, prompting organizations to address care deficiencies revealed by public data [73]. Existing studies highlight the need for incentive policies, which could effectively enhance quality competition in the healthcare market. Policy making should focus on how these incentives can further encourage high-performing hospitals to improve healthcare outcomes. In this way, a patient-centered public reporting system can be established, enabling the timely collection of patient experiences and outcomes, promoting healthcare transparency.

Second, many practitioners lack formal training in medical coding or documentation improvement, and adherence to standards for physician queries and medical record corrections is suboptimal, with numerous institutions failing to follow prescribed timelines and correction procedures (Crews et al., 2016; Lagu et al., 2019; Martin M et al., 2013; Rhoads et al., 2009). By leveraging data for quality improvements, engaging healthcare staff in improvement activities, and clarifying roles and goals, these strategies indirectly contribute to achieving public reporting's quality improvement objectives (Davila et al., 2024).

Third, developer should consider the different stakeholders when developing the system, most public reporting system are directed toward organizational use, patients make up the largest group using public reporting systems, family members of patients also participate in these systems, system development should account for the needs of patient families, particularly in data-sharing options.

6 Conclusions and limitations

In this paper, a systematic review has been conducted of the literature based on the sample of 60 review. The rise of patient-centered public reporting and user-generated data has expanded the scope of public reporting beyond patients, providers, and regulators to include analysts, monitoring systems, and digital health experts. This shift reflects the broader impact of digitalization in healthcare information dissemination and decision-making.

This study provides an up-to-date and comprehensive review of public reporting in healthcare, identifying key trends, challenges, and future research directions. The findings highlight the importance of standardization, usability, and digital transformation in public reporting. Addressing these challenges can enhance data transparency, stakeholder engagement, and healthcare quality outcomes. Future research should explore interdisciplinary approaches, stakeholder behaviors, and technological advancements to further improve public reporting systems in the healthcare sector.

This systematic literature review has certain limitations. First, as the study focuses specifically on public reporting in healthcare, the search strategy only included studies that explicitly mentioned public reporting. Consequently, relevant research exploring similar concepts without using this exact term may have been overlooked. Additionally, the review only included studies published in English, potentially omitting valuable insights from research in other languages.

Second, the process of coding the literature was conducted manually, which introduces a risk of subjectivity despite efforts to minimize bias through multiple rounds of coding and verification. Third, while this study categorizes public reporting within individual, organizational, and system-level frameworks, we acknowledge that alternative classifications may also provide valuable insights.

This systematic literature review examines public reporting from a global perspective, analyzing its evolution, impact, and future potential. By exploring the relationships between public reporting systems, data components, and key stakeholders, this study provides valuable insights into the effectiveness of public reporting in driving healthcare transparency and quality enhancement.

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